DSB300 Projects

This document contains all the projects on offer in DSB300 Data Science Capstone Project in semester 2 2026. Each project has a title with the following formatting:

<industry partner name>: <short title of project> (<number of teams who can take this project> Team)

Projects are supervised by the industry partner, listed directly below each project title, and the academic team in DSB300. Here, an overview with some details of each project is provided.

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The Partner Organisations

The **Academy of Financial Services (AFS)** is a not-for-profit global membership association of academics, students and practitioners in the field of Personal Financial Planning. Since 1991, AFS has published a peer-reviewed academic journal called *Financial Services Review* (FSR).

Established in 1991, the **Queensland Academy of Sport (QAS)** is the Queensland Government's high performance sports agency responsible for preparing Queensland elite athletes, teams and coaches for world class success.

The **Cotton Research and Development Corporation (CRDC)** is one of Australia's 15 Research and Development Corporations, dedicated to agricultural innovation. Focused on sustainable cotton production, CRDC has invested over $420m since 1990 into research, development and extension (RD&E) to enhance the economic, social, and environmental contribution of cotton.

The **Royal Automobile Club of Queensland (RACQ)** is mutual business, owned by over 1.7 million members. It provides services in roadside assistance, motoring services, insurance, banking, travel and solar products. RACQ is proud to be the independent advocate for its members and all Queenslanders on the roads, in the home and within our communities. It advocates for changes that make our state safer, more affordable and more sustainable.

Ubrix: We integrate connected infrastructure and AI-driven insights to transform how energy is used across buildings and cities — delivering measurable improvements in efficiency, performance, and sustainability.

QUT Sport offers opportunities for students, staff and members of the community to engage in a range of physical activities and programs.

Academy of Financial Services (AFS): Data Interfaces (1 Team)

Partner Supervisors: Shawn Brayman, Dr Norah Feng

Project Description with Academy of Financial Services (AFS) and Queensland University of Technology

Data interfaces for pulling, cleaning and populating journal paper metadata

The Academy of Financial Services (AFS) is a not-for-profit global membership association of academics, students and practitioners in the field of Personal Financial Planning. Since 1991, AFS has published a peer-reviewed academic journal called *Financial Services Review* (FSR). In 2023, FSR was transformed into a Diamond Open Access journal (no fees to access, submit or have research published) and hosted on a technology platform called Open Journal System (OJS). OJS is an open-source platform developed at Simon Fraser University, Canada that currently hosts 58,000 journals globally. As an open-source platform, OJS is well documented with multiple interfaces/plugins and import/export capabilities. Over the past 3 years, all past issues of FSR have been manually loaded by volunteers into our OJS instance, hosted by the University of Georgia, including critical metadata and PDF galleys, making the over 750 research submissions electronically accessible.

The project with QUT includes some key objectives:

- Intelligent electronic review of existing metadata to isolate “bad data”, and if possible, to clean up and refresh the metadata. Bad data can be:
    - unreadable characters copied from the PDF copies;
    - Lost spaces between words (i.e. The UniversityofNorthCarolinaat Charlotte,Charlotte,NC)
    - Institution footnote reference attached to author names
- A significant number of PDF galleys are image only PDFs. Look for effective ways to transform these into machine readable PDFs so the content is accessible. Previous attempts failed due to formulas or poor quality photo copies.
- Some important meta data has not been uploaded (ie references/citations). Tools exist with CrossRef, a global DOI registration body, to scan reference data and match their global database to generate matches and DOI “addresses”. Can we automate this process to add pass the reference data to the CrossRef tool and then load meta data with DOI references back into OJS?

Since OJS is a globally utilized platform, skills and knowledge gained in this project may be practically transferable to other opportunities in the future.

Shawn Brayman
Immediate Past President
Academy of Financial Services

Academy of Financial Services (AFS): Journal Citation Analytics (1 Team)

Partner Supervisors: Shawn Brayman, Dr Norah Feng

QUT Data Science Capstone DSB300 Project Overview

Industry Partner: Academy of Financial Services (AFS)

Journal citation analytics: How can Financial Services Review (FSR) attract more citations?

About the Academy. The Academy of Financial Services (AFS) is a not-for-profit global membership association of academics, students and practitioners in the field of Personal Financial Planning. Since 1991, AFS has published Financial Services Review (FSR), a peer-reviewed academic journal called. In 2023, FSR was transformed into a Diamond Open Access journal (no fees to access, submit or have research published) and hosted on a technology platform called Open Journal System (OJS). Over the past 3 years, all past issues of FSR have been manually loaded by volunteers into our OJS instance, hosted by the University of Georgia, including critical metadata and PDF galleys, making the over 750 research submissions electronically accessible. FSR is listed on SCOPUS and listed in the following academic rankings:

- Chartered Association of Business Schools (CABS) Academic Journal Guide
- ABDC Journal Quality List - Australian Business Deans Council
- Cabells Scholarly Analytics

FSR's standing in academic rankings is driven largely by citation metrics, and stronger rankings attract higher-quality submissions. The editorial team, therefore, is seeking your support to develop data-driven insights, addressing one central question:

How can FSR attract more citations?

This overarching research question can be approached from several complementary angles. The bibliometric literature shows that many article-level features correlate with citation counts, but the strength, and even the direction, of these effects varies considerably by field, time period and journal, and the evidence is rarely causal (Kousha and Thelwall, 2024). That is exactly why FSR is worth studying on its own terms, rather than assuming that findings from other disciplines carry over to personal financial planning.

The output should give the editorial team practical, evidence-based guidance: which kinds of papers, authors and topics tend to attract citations in this field, and how soon impact can reasonably be judged.

Data

Students will work with Scopus metadata for FSR and compare it against its two closest competitors:

- **Financial Planning Review** (Wiley, sponsored by the CFP Board)
- **Journal of Financial Counseling and Planning** (Association for Financial Counseling and Planning Education)

Guiding questions

We have prepared several guiding questions to assist student teams with their thinking on how to best approach this problem.

Question A: Do "star" authors drive citations?

- **Question:** Are citations concentrated among a small number of prolific or well-known authors, and does an author's prior track record predict how their FSR papers perform?

- **Approach:** Build an author-level dataset, measure citation concentration (for example, the share of total citations held by the top authors), and test whether prior publication and citation counts predict the performance of later papers.
- **Context:** Author and collaboration factors are among the more consistent predictors of citations in the literature, though they are far from deterministic (Kousha and Thelwall, 2024).

Question B: Which topics and keywords attract citations?

- **Question:** Do certain keywords, topics or title features associate with higher citation counts?
- **Approach:** Text-mine titles, abstracts and author keywords. Use term frequencies and topic modelling, then compare high-cited and low-cited clusters. Comparing FSR's topic mix to its competitors can also reveal high-citation areas that FSR is currently under-serving.

Question C: Does a clearly written abstract get cited more?

- **Question:** Does abstract readability, measured with the Gunning Fog Index, relate to citations in FSR?
- **Why this is interesting:** The evidence is genuinely mixed, which makes finance a worthwhile test case. Some studies report that more readable abstracts attract more citations (Gazni, 2011; Didegah and Thelwall, 2013), while others find the opposite, with more complex abstracts attracting more citations (Sienkiewicz and Altmann, 2016). The answer appears to depend on the field, so it is an open question for personal financial planning.
- **Approach:** Compute the Fog index (and one or two alternative readability measures) for each abstract, then model citations against readability while controlling for paper age, topic and author.

Question D: When do citations peak, and which papers become "high hitters"?

- **Question:** How many years after publication do FSR papers typically peak, and can the papers that go on to become highly cited be spotted early?
- **Context:** The typical pattern is a citation peak a few years after publication followed by decline, and the large majority of papers decline after their maximum (Ke, Ferrara, Radicchi and Flammini, 2015). A minority follow other patterns: "sleeping beauties" gain recognition only after a long delay, while "flash in the pan" papers are noticed quickly but fade. Distinguishing papers with lasting impact from short-lived ones matters for how soon a paper's success can fairly be judged.
- **Approach:** Build a yearly citation curve for each paper, identify peak timing, fit a simple ageing curve, and look for early indicators that flag eventual high-hitters.

Tools and getting started

Suggested starting points:

R

- quanteda: text wrangling, tokenisation and document handling.
- quanteda.textstats: readability measures via textstat_readability(), which includes the Gunning Fog Index, plus lexical-diversity and term statistics. This is the most direct route for Question C.
- tidyverse for general data wrangling and plotting.
- seededlda or stm for topic modelling (Question B).

Python

- pandas for data wrangling.
- textstat for readability, including a gunning_fog function (Question C).
- scikit-learn, gensim or BERTopic for topic modelling (Question B).
- nltk or spaCy for tokenisation and text cleaning.

Data access

- Scopus CSV or RIS exports for FSR and the two competitors are the primary source.
- pybliometrics (Python) can automate Scopus API pulls if API access is available.
- OpenAlex and Crossref are free options for supplementary or cross-checked citation data.

Suggested readings

- Didegah, F., & Thelwall, M. (2013). Which factors help authors produce the highest impact research? Collaboration, journal and document properties. *Journal of informetrics*, *7*(4), 861-873. https://doi.org/10.1016/j.joi.2013.08.006.
- Gazni, A. (2011). Are the abstracts of high impact articles more readable? Investigating the evidence from top research institutions in the world. *Journal of Information Science*, *37*(3), 273-281. https://doi.org/10.1177/0165551511401658.
- Ke, Q., Ferrara, E., Radicchi, F., & Flammini, A. (2015). Defining and identifying sleeping beauties in science. *Proceedings of the National Academy of Sciences* 112(24): 7426-7431. https://doi.org/10.1073/pnas.1424329112.
- Kousha, K., & Thelwall, M. (2024). Factors associating with or predicting more cited or higher quality journal articles: An Annual Review of Information Science and Technology (ARIST) paper. *Journal of the Association for Information Science and Technology* 75(3): 215-244. https://doi.org/10.1002/asi.24810.
- Sienkiewicz, J., & Altmann, E. G. (2016). Impact of lexical and sentiment factors on the popularity of scientific papers. *Royal Society Open Science*, *3*(6). https://doi.org/10.1098/rsos.160140.

The methods in this project transfer well beyond academic publishing. You will gain hands-on experience with text mining and natural language processing on a real text corpus, in addition to citation and metadata APIs (Scopus, OpenAlex, Crossref), and with building reproducible analysis pipelines in R or Python. You will practise turning a messy dataset into clear, evidence-based recommendations for a real stakeholder. These are the same skills used in roles such as marketing and customer-text analytics, competitive and market intelligence, and research or R&D analytics, and they apply to any data work that involves large volumes of unstructured text.

Queensland Academy of Sport (QAS): YouFor2032 Analytics (2 Teams)

Partner Supervisors: Matt Glossop

TALENT IDENTIFICATION PROGRAM – PROJECT 1A & 1B

TITLE

Analysis, Visualisation and Presentation of YouFor2032 Data to Sports.

DESCRIPTION

The QAS Youfor2032 talent search program identifies and supports athletes who show extraordinary potential to achieve medal success at the Brisbane 2032 Games. These athletes are typically coming from outside of the traditional sport talent pathways. We also offer sport-specific talent identification opportunities in early specialization sports including diving, skateboarding, and BMX freestyle. Our commitment lies in maximizing the potential of every athlete, whether they qualify for the Youfor2032 program or not.

OBJECTIVES

Each year, the QAS Talent Team travels throughout the state to capture athletic testing data for aspiring world class athletes. This information is then transferred to coaches, who use it to identify athletes most likely to succeed in their sport. This specific interaction is the basis for the project.

Objective 1 – Review approaches for visualising the data and rankings of athletes given coach criteria

Objective 2 – Review approaches for capturing expert criteria systematically? (e.g. multi-criteria decision making, analytic hierarchy process)

Objective 3 - Develop a unique method for ranking athletes and making trade-offs between criteria such as through pairwise comparisons (e.g. Analytic Hierarchy Process)

Objective 4 – GROUP 1A	Objective 4 – GROUP 1B
Develop a dashboard to enable interactive visualisation of the data, rankings and user choices and updates to their criteria	Develop a system for adjusting an athletes score (or rating), based on relative age (eg. Birth month) and maturation age (eg. Biological development)



DATA

Data is available in .xls format, approximately 6000 athletes over 4 years.

Samples of how data is currently presented to sport are available

This data includes appropriate data to calculate relative and maturation age metrics.

EXAMPLE OF CURRENT SPORT SELECTION CRITERIA	
FILTERS (IN OR OUT)	**WEIGHTINGS (vs Age & Gender)**
Female Only Age 15-17 Location: SEQ Only Background: Team Sport	Height – 15% Predicted Height – 10% Sprint – 25% Vertical Jump – 15% Co-ordination Test – 10% Upper Body Power – 10% Endurance – 15%

Queensland Academy of Sport (QAS): Coach Projection Analytics (2 Teams)

Partner Supervisors: Matt Glossop



TALENT IDENTIFICATION PROGRAM – PROJECT 2

TITLE

The success of early coach projections in the YouFor2032 Talent Search

DESCRIPTION

The QAS Youfor2032 talent search program identifies and supports athletes who show extraordinary potential to achieve medal success at the Brisbane 2032 Games. These athletes are typically coming from outside of the traditional sport talent pathways. We also offer sport-specific talent identification opportunities in early specialization sports including diving, skateboarding, and BMX freestyle. Our commitment lies in maximizing the potential of every athlete, whether they qualify for the Youfor2032 program or not.

OBJECTIVES

Coaches are required to select athletes based purely on athletic testing and demographic data, having not seen the athlete perform any sports specific skills or movements. During this selection process, they are required to allocate a rating of likelihood the athlete will succeed in the program;

Alpha- a strong selection which meets all criteria

Beta- a good candidate who may be deficient in one or two non-critical areas

Smokey- an athlete who does not match the ideal prototype, but has ‘something’ that makes them worth a selection

Objective 1: develop conditional models for alpha/beta/smoky athletes and analyse the rates / chances of progression to phase 1, 2 and 3. Can coaches accurately predict those who would succeed, or is it random? Which sport/types of sports have a good hit rate?

Objective 1a. identify covariates based on the literature and the data (variable selection) that support this analysis

Objective 2. Analyse the effect of age on selection and progression. Are selections more stable as age increases?

Objective 3. Analyse the effect of maturation age on selection and progression and compare with age effects. Eg. Are smokey selections correlated with or explained by biological/maturation age?

DATA

Data is available in .xls format, approximately 6000 athletes over 4 years.

Progression is characterised by level reached within the program, with selection/deselection timepoints spread longitudinally across 2 years.

This data includes appropriate data to calculate relative and maturation age metrics.



Cotton Research Development Corporation (CRDC): Survey Analytics (2 Teams)

Partner Supervisors: Dr Burton Wu

Title: Transforming Survey Data for Analysis and Reporting

Project Summary

Build a scalable data pipeline that transforms fragmented multi-year grower survey data into a structured, analysable database for reporting, benchmarking, and decision-making.

Project Overview

There are two projects based on the same underlying work, both covering the core set of variables, with an additional focus layer—water (GPWUI) or nutrient (kg N/ha): the first is analysis-led (Python/statistics), while the second centres on PowerBI dashboarding (slice-and-dice) and region/segment (benchmarking) reporting.

Objective (What)

- Develop a scalable data solution that transforms CRDC's annual grower survey data into a structured, analysable dataset.
- The project will:
  - Standardise survey data across multiple years despite changing question formats
  - Build a clean, consistent data model suitable for a data warehouse
  - Enable longitudinal analysis of key variables (e.g. trends, behaviour changes, adoption patterns)
  - Support future statistical analysis (e.g. confidence intervals, trend modelling)
- This project sits at the intersection of data engineering and data science, offering students exposure to real-world messy data challenges.

Problem (Why)

- CRDC collects rich survey data every year, but it is difficult to analyse over time because:
  - Question numbers and wording change between surveys
  - Equivalent questions don't align neatly across years
  - Response options evolve or differ in format
  - Data is stored in spreadsheets without a consistent structure
- As a result, valuable insights remain locked in fragmented datasets, limiting the ability to track trends or support evidence-based decision-making.
- This project tackles a real-world data challenge: how to impose structure and consistency on messy, evolving survey data.

End User

- Data managers responsible for maintaining a reliable data warehouse with consistent, structured survey data to support accurate reporting and analysis.

Technical Requirements

- Python for data cleaning, transformation and analysis.
- SQL for database design and querying.
- Excel for data exploration and validation.
- Power BI (dashboard project) for reporting and visualisation.

Project Deliverables (Phases)

- **Phase 1: Data Mapping & Understanding**
  - Explore datasets across multiple years
  - Identify and map equivalent questions and responses
  - Define a consistent structure for key variables
- **Phase 2: Data Transformation (Python)**
  - Clean and standardise survey data
  - Build reusable scripts to align questions and responses across years
  - Convert raw files into a unified dataset
- **Phase 3: Data Model & Storage (SQL)**
  - Design a structured database (warehouse-style)
  - Load transformed data for easy querying and analysis
  - Enable cross-year comparisons
- **Phase 4: Insights & Analysis (Python or PowerBI)**
  - Perform trend analysis and apply basic statistical methods using the structured master database
  - Leverage the unified dataset to slice and dice results (e.g. by region, grower segment, or time)
  - Generate reliable, comparable insights on grower behaviour and industry trends across years

What Students Will Learn

- ETL development using Python and SQL.
- Database design and data warehousing fundamentals.
- Cleaning, standardising and integrating messy real-world survey data.
- Data quality, validation and governance practices.
- Trend analysis, benchmarking, reporting and dashboard development.
- Converting fragmented spreadsheets into a structured database for analysis and decision-making.

Future Extensions

- Extend the project to include analysis and reporting on open-ended (free-text) survey responses.

Cotton Research Development Corporation (CRDC): AI Metadata Pipeline (2 Teams)

Partner Supervisors: Dr Burton Wu

Title: Turning Scientific Papers into Searchable Knowledge

Project Summary

Build a modern data pipeline that turns research papers into structured, queryable evidence — extracting both metadata and key quantitative findings for analysis and decision-making.

Project Overview

This project should follow a more traditional Python-first ETL and data engineering approach, while incorporating LLMs selectively for semantic extraction, contextual interpretation, and difficult document understanding tasks. This project can be spined out into two groups: one focused on text-based extraction and another focused on tables/images-based extraction. Access to local LLMs potentially beneficial for iterative high-volume processing and experimentation.

Objective (What)

- Develop an AI-assisted pipeline that transforms unstructured research PDFs into a **structured, queryable** dataset by extracting both metadata and quantitative findings.
- The system should not only extract fields, but also perform **limited contextual interpretation**, including:
    - identifying variables, values, units, and conditions
    - capturing explicitly stated limitations or assumptions
    - normalising inconsistent representations across documents
- This project focuses on **document-level understanding and structuring**, not cross-document synthesis.
- The project will:
    - Extract and standardise document-level metadata (e.g. title, authors, year, source)
    - Identify and extract key quantitative findings (statistics) from research papers
    - Capture contextual information such as variables, units, and conditions
    - Preserve provenance (e.g. page number, source text) for traceability
    - Store outputs in a structured format suitable for downstream analysis
- All outputs must be clearly defined, contextualised, and **strictly grounded in source text with full traceability** (e.g. page references and excerpts).
- This project extends traditional ETL pipelines to **unstructured documents**, introducing modern data science techniques for knowledge extraction.

Problem (Why)

- A large amount of valuable information exists in research papers, but it is difficult to extract and use systematically.
- Key challenges include:
    - Important findings are embedded within long text, tables, and figures
    - Statistics are presented inconsistently across papers
    - Context (e.g. conditions, limitations) is often unclear or implicit
    - Metadata is incomplete or poorly standardised
- As a result:
    - Data extraction is manual and time-consuming

    - Cross-document comparison is difficult
    - Valuable insights remain unstructured and underutilised
- This project addresses a key challenge: **How can we systematically extract structured, reliable evidence from unstructured research documents while preserving context and traceability?**

End Users

- Data managers responsible for maintaining a structured evidence warehouse (metadata + statistics) to support analysis, decision-making, and AI-enabled knowledge systems (e.g. RAG (retrieval-augmented generation)).

Technical Requirements

- **Environment**: Desktop-based; Python used for workflow orchestration
- **Tools**: Agentic coding environments (e.g. Google Antigravity, Claude Code)
- **Processing**: LLM-based extraction using structured prompts and templates
- **Model Choice**: Performance should be improved through workflow design, structuring, and validation rather than relying on a single high-performance model; experimentation with different models is encouraged
- **Workflow Design**: Modular ETL-style pipeline with reusable components (e.g. subagents, skills, hooks)
- **Data Scope**: Closed corpus only (no external data retrieval)
- **Validation & Control**: Outputs must be structured, consistent, and traceable to source documents
- **Version Control**: Git-based, demonstrating progression and iteration
- **Outputs**: Structured dataset (metadata + statistics) with provenance

Project Deliverables (Phases)

- **Phase 1: Data Understanding & Schema Design**
    - Review sample PDFs
    - Define:
        - metadata schema
        - statistics extraction schema
    - Define what qualifies as a “statistic”
    - Define normalisation and provenance requirements
- **Phase 2: Metadata Extraction Pipeline**
    - Extract and standardise document-level metadata
    - Store outputs in structured, schema-compliant format
- **Phase 3: Statistics Extraction Pipeline**
    - Identify and extract quantitative findings
    - Capture context (variable, value, unit, conditions)
    - Preserve source text and page references
    - Apply validation and normalisation where appropriate
- **Phase 4: Data Integration & Structuring**
    - Combine metadata and statistics into a unified dataset
    - Ensure consistency, traceability, and schema compliance
- **Phase 5: Validation & Evaluation**
    - Evaluate extraction accuracy on a labelled sample
    - Assess:
        - correctness
        - completeness
        - format adherence

        - consistency and traceability
    - Identify common errors (e.g. incorrect context, duplication, missing fields, normalisation inconsistencies)

What Students Will Learn

- Designing **Python-based data pipelines for unstructured data**
- Applying ETL concepts to text and documents
- Extracting structured information using LLMs
- Building systems with traceability, validation, and schema control
- Evaluating extraction quality and handling ambiguity

Cotton Research Development Corporation (CRDC): AI Report Synthesis (2 Teams)

Partner Supervisors: Dr Burton Wu

Title: Build an Agentic AI Research Analyst: From Research Papers to Structured Reports

Project Summary

Build an agentic AI workflow that replicates Deep Research–style systems — turning a set of research papers into a structured, high-quality report, first rapidly, then as a controlled and reproducible system.

Project Overview

This project should be approached as a modern agentic AI workflow, with LLMs performing most of the reasoning, synthesis, and orchestration tasks while Python provides lightweight workflow control, validation, and reproducibility. The focus should be on rapid prototyping, modular subagents/skills, structured prompting, and evidence-grounded synthesis pipelines rather than traditional software-heavy development.

Objective (What)

- Develop an AI-assisted system that transforms a fixed set of supplied research papers into a **structured, detailed, and evidence-grounded report**.
- The project will:
    - Process multiple research papers (e.g. five PDFs) on a defined topic
    - Extract and organise key elements such as findings, methods, and limitations
    - Generate a structured synthesis report using a predefined template
    - Progress from a rapid prototype to a **more engineered and reproducible workflow**
- The workflow mirrors the structure of modern **deep research systems** (e.g. multi-step synthesis pipelines), adapted to operate on a fixed set of supplied documents.
- The final output should resemble a **structured research briefing**, including:
    - comparisons across studies
    - key findings and supporting evidence
    - identified trends and gaps
    - grounded conclusions
- This project introduces students to **modern data science applied to unstructured data**, with a focus on **agentic workflows** and **system design**.

Problem (Why)

- Organisations often need to turn a small set of reports or research papers into a clear, detailed briefing. However:
    - Key insights are spread across long and complex documents
    - Similar ideas are described differently across sources
    - Manual synthesis is slow, inconsistent, and difficult to standardise
    - Naive AI outputs are often shallow, unstructured, or poorly grounded
- In this project, the task is not to search for new information, but to:
    - Work within a fixed set of documents
    - Produce structured, non-random, detailed outputs
    - Ensure results are grounded in source material

- This project addresses a key challenge: **How can we systematically synthesise existing research outputs into a structured, reliable report using AI workflows — to support comparison and better-informed research and investment decisions?**

End Users

- R&D managers seeking structured insights from research papers to identify gaps and prioritise investment opportunities.

Technical Requirements

- **Environment**: Desktop-based; Python used for workflow orchestration
- **Tools**: Agentic coding environments (e.g. Google Antigravity, Claude Code)
- **Processing**: LLM-based extraction and synthesis using structured prompts and templates
- **Storage**: Vector database (i.e. embedding) may be required
- **Model Choice**: Performance should be improved through workflow design, structuring, and validation rather than relying on a single high-performance model; experimentation with different models is encouraged
- **Workflow Design**: Modular ETL-style pipeline with reusable components (e.g. subagents, skills, hooks)
- **Data Scope**: Closed corpus only (no external data retrieval)
- **Validation & Control**: Outputs must be structured, consistent, and traceable to source documents
- **Version Control**: Git-based, demonstrating progression and iteration
- **Outputs**: Structured reports and intermediate artefacts (e.g. summaries, evidence tables, comparison matrices)

Project Deliverables (Phases)

- **Phase 1: Rapid Prototype**
    - Build an initial workflow using agentic coding tools
    - Generate a full synthesis-style report from the supplied papers
    - Focus on rapid experimentation, speed, and demonstrating feasibility
- **Phase 2: Workflow Structuring**
    - Break down the workflow into clear modular steps:
        - per-paper summaries
        - extraction of key elements
        - structured report sections
        - evidence tables and comparison matrices
    - Define a consistent report template
    - Introduce structured intermediate artefacts (e.g. paper summaries, evidence tables, comparison matrices) to improve grounding, traceability, and controllability
- **Phase 3: Engineering & Refinement**
    - Convert the workflow into a more structured, reusable, and reproducible pipeline:
        - modular prompts or skills
        - reusable workflow components
        - consistent structured output formats
    - Improve consistency, grounding, clarity, and structure of the report
    - Reduce comm LLM failure modes through workflow design and validation
- **Phase 4: Validation & Evaluation**
    - Ensure report content is grounded in the supplied papers
    - Check consistency, completeness, and evidence coverage of outputs

- Compare early prototype vs refined workflow
- Identify common failure modes (e.g. unsupported claims, hallucinations, repetition, shallow analysis)

What Students Will Learn

- Designing modular **LLM-based workflows**, not just prompts
- Moving from **prototype → engineered and reproducible system**
- Structuring unstructured data into controlled and usable outputs
- Controlling AI outputs using templates, decomposition, and workflow orchestration
- Evaluating grounding, quality, and consistency in non-deterministic systems

Understanding trade-offs between flexibility, automation, and reproducibility in modern AI workflows

Royal Automobile Club of Queensland (RACQ): Road Safety Analytics (1 Team)

Partner Supervisors: Dr Ian Jeffreys

Understanding Fatal Pedestrian Crashes in Australia (Queensland)

Background and Rationale

Pedestrian fatalities remain a critical road safety concern in Australia. For example, in Queensland between 2016 and 2024 (9 years), a troubling number of fatal pedestrian collisions occurred (n=263). A further 29 have occurred to October 28, 2025.

Preliminary analysis of Queensland data (2019-23) indicates that many pedestrian deaths occur away from intersections at midblock locations and in zones with varying speed limits. Other data suggests that many victims were engaged in non-crossing activities (e.g., standing, lying, or walking along the carriageway), and that infrastructure gaps, behavioural factors, speed environments and vehicle/driver factors may be contributing to these tragedies.

Research data

Data to include:

- Reported road traffic crashes Serious crashes attributed with a traffic violation, within Queensland (Data from QTMR requested by RACQ)
- Road Crash Locations, Qld Government Open Data Portal. https://www.data.qld.gov.au/dataset/crash-data-from-queensland-roads/resource/e88943c0-5968-4972-a15f-38e120d72ec0

Possible date points include:

1. Crash locations (lats and lons)
2. Crash description
3. Injuries and fatalities
4. Traffic violations (Speeding, Exceeded Blood Alcohol Limit, etc)
5. Aggregate vehicle movement/actions

Objectives

This proposal seeks to improve understanding of the causes, patterns, and contributing circumstances of pedestrian road deaths and serious injuries.

Data is provided in two sets that first will require linking, before modeling the relative impacts on the number of crashes arising from: types of accidents, cause of accident, location of accident.

The goal is to help inform evidence-based interventions, such as calls for no-blame investigations to further explain, target and reduce future pedestrian deaths and serious injuries, and help towards achieving Australia's Vision Zero target.

Royal Automobile Club of Queensland (RACQ): Petrol Market Analytics (1 Team)

Partner Supervisors: Dr Ian Jeffreys

Understanding Petrol Markets in Greater Brisbane

Background and Rationale

This project will investigate pricing dynamics in the Regular Unleaded Petrol retail market in Greater Brisbane.

Regular Unleaded Petrol (RULP) is the most common petrol garde sold in retail petrol markets and the most common liquid fuel used in passenger cars.

Retail prices in the Greater Brisbane Regular Unleaded Petrol (RULP) market usually follow an Edgeworth Cycle. This is, a saw tooth cycle, typified by a sharp increase in price or a price hike, followed by a slow decrease or discounting phase. During the price increase phase individual retail sites jump from a price close to wholesale, to a new high price typically 50 cents above wholesale. In the discount phase rice then begin to slowly fall as each site progressively undercuts their neighbours.

A similar price cycle is observed on the Gold Coast, Sunshine Coast. In the other states, the Sydney and Melbourne cycles are very similar to Brisbane. The average price in these four centres is also similar. Adelaide has shorter cycle, with greater number of days of relatively cheap pricing. Perth has a weekly cycle with Tuesday being the cheapest day and Thursday the dearest. Average prices in Perth and Adelaide tend to be 10 cents cheaper than in Brisbane. Canberra, Hobert, Darwin and regional Australian markets do not exhibit price cycle behaviour.

The different brands in Brisbane tend to behave differently. Typically, brands can be classified in to three broad groups:

- Group A – the price leaders, those brands most likely to initiate a price hike, and often the higher priced brands.
- Group B – the price follower, those brands that rarely imitate a hike but will quickly follow the lead of others, and of the mid-priced brands.
- Group C – the discounters, those brands that aggressively discount their prices, these brands tend to be the slowest to hike and the quickest to discount. On average these brands are the cheapest in the market.

Research data

Data to include:

- Retail price data – an hourly time step data set, for all fuel types at all retail site in Queensland, including:
  - Retail price data – hourly
  - All fuel types, Diesel, RULP, E10, PULP95 and PULP98
  - May 2022 to date
  - Site branding
  - Site location address, and lats and lons
- Wholesale price data, including:
  - Average Terminal Gate Price (the wholesale price) – set and reported daily
  - Brisbane terminals
  - All fuel types, Diesel, RULP, E10, PULP95 and PULP98
  - May 2022 to date
- These datasets are collated by RACQ using public and proprietary sources

Objectives

This proposal seeks to model the nature of price cycle in the Greater Brisbane region. Measures could include the number of peaks in a year, the amount of time in the expensive periods (when the retail price exceeds TGP plus 35 cents) and the amount of time in cheap periods (when the retail price falls below TGP plus 7 cents).

In relation to the above, the project will also identify differences in behaviour between different fuel station brands, brand groupings, and different suburbs. It may extend to investigation of other pricing metrics and pricing behaviours such as the likelihood of initiating a price hike and/or price discounting.

Ubrix: Hotel Occupancy Analytics (2 Teams)

Partner Supervisors: Nick Czeperko (Ubrix)

AI Driven Operational Optimisation for Quest River Park Hotel

Ubrix Partner: Quest River Park Central, 120 Mary Street, Brisbane, Qld 4000

Project Overview

Quest River Park Central, 120. Mary Street Brisbane Qld 4000 will partner with QUT to host a data science team for the full 13-week capstone project. This project focuses on identifying the key determinants of occupancy — specifically the relative influence of pricing, online visibility and exposure, and guest satisfaction ratings — to enable evidence-based revenue strategy.

Hotels operate in a dynamic environment where room rate, search ranking, review scores, and competitor behaviour all interact to drive occupancy. Understanding the relative weight of each factor — and how that weight shifts across seasons, room types, and booking channels — is a high-value analytical problem that is well-suited to data science methods.

The outputs will give Quest River Park Central a quantitative basis for revenue management decisions: where to prioritise investment (pricing strategy, review response, OTA optimisation), and what occupancy to forecast under different scenarios.

Core Research Questions

The student team will aim to answer the following questions:

- What is the relative influence of room rate, online review score, and OTA visibility on occupancy rate?
- Is there a price elasticity threshold — a rate above which occupancy drops sharply — and does it differ by guest segment or booking channel?
- Can a predictive model be built to forecast occupancy given a set of pricing, visibility, and sentiment inputs, and how accurately does it perform out-of-sample?
- Extension questions:
    - How does relative influence vary by season and room type?
    - How does review score (overall and by sub-dimension: cleanliness, location, value, service) correlate with occupancy, and is there a minimum score below which bookings decline materially?
    - Which OTA channels (Booking.com, Expedia, direct, etc.) drive the most bookings, and how does search ranking or listing quality on each platform relate to conversion?
    - Are there lead-time effects — do certain pricing or visibility factors matter more at 30/60/90 days out than at last-minute booking windows?
    - What scenario analysis can be offered — e.g. “if we raise ADR by 10% and our review score improves from 8.2 to 8.6, what happens to projected occupancy?”

Data Requirements

For the project to be viable, Quest River Park Central will need to provide internal data, and the team will supplement this with publicly available external data. The minimum viable dataset is categories 1 and 2. Categories 3 and 4 significantly enrich the analysis.

1 — Reservation & occupancy data (internal)
Daily occupancy rate and rooms available, minimum 2–3 years of history
Average daily rate (ADR) and revenue per available room (RevPAR) by day and room type
Booking channel per reservation (direct, Booking.com, Expedia, corporate, walk-in, etc.)
Booking lead time per reservation (days between booking date and check-in)
Length of stay per reservation
Cancellation rate by channel and lead time if available

2 — Online review & rating data (internal + public)
Historical review scores from Booking.com, Google, TripAdvisor, and Expedia — overall and sub-scores (cleanliness, value, location, service) with dates
Volume of reviews per platform per month
Quest's own guest satisfaction survey results if available (NPS, CSAT, or equivalent)
Response rate to reviews and average response time if tracked

3 — Online visibility & channel data (internal + public)
OTA listing rank or position history if available from channel manager exports
Click-through and conversion data from OTA extranets (Booking.com Partner Hub, Expedia Partner Central) if accessible
Direct website traffic and conversion data from Google Analytics or equivalent
Promotion and discount history (flash sales, loyalty rates, last-minute deals) with dates
Any channel manager data exports (e.g. SiteMinder, RateGain) covering rate parity and visibility

4 — Competitor & market context (largely public)
Competitor rate benchmarking data if Quest subscribes to a tool (e.g. OTA Insight, RateShopper) — even a partial export is useful
STR or AHS market occupancy benchmarks for the Brisbane CBD apartment-hotel segment if available
Local event calendar for Brisbane (conferences, festivals, public holidays) — the team can source this externally
Macroeconomic indicators (interstate travel volumes, ABS visitor nights) — the team can source these from Tourism Research Australia

Indicative Analytical Approach

While the student team will determine their own methodology, the following approaches are well-suited to this problem and within scope for a penultimate-year data science cohort:

- Regression modelling (OLS, ridge, or lasso) to quantify the marginal effect ofeach determinant on occupancy, with controls for seasonality and day-of-week
- Feature importance analysis (random forest or gradient boosting) to rankpredictors and identify non-linear threshold effects
- Sentiment analysis of review text (if full review text is available) to extract sub-themes that predict score changes
- Time-series decomposition to separate trend, seasonal, and residual components before modelling
- Scenario simulation tool or dashboard allowing Quest staff to input rate and review assumptions and view projected occupancy

QUT Sport: Business Analytics (2 Teams)

Partner Supervisors: Joseph Hayden

Title: Designing enhanced operational and financial reporting in Microsoft Excel for QUT Sport Fitness & Aquatic Centres

Overview

QUT Sport seeks to simplify and improve its monthly operational and financial reporting for the on-campus Fitness & Aquatic Centres. The current reporting process relies on manual Excel workflows, raw report exports, and reconciliation against existing internal reporting outputs. This project will focus on building an enhanced Microsoft Excel reporting capability that is practical, maintainable, and useful for QUT Sport staff.

Students will design a structured Excel reporting workbook or workbook system that can ingest supplied raw report exports, clean and combine the data, and present clear reporting outputs for memberships, financial performance, visitation, usage trends, and peak-time analysis. The solution should remove inaccuracies associated with the current manual reporting process and support better decision-making while avoiding unnecessary technical complexity for staff who currently work primarily in Excel.

Objectives

- Map QUT Sport's existing manual reporting workflow and identify where Excel-based automation can reduce repeated manual reporting.
- Use supplied raw report exports to build repeatable Excel data preparation steps, including cleaning, categorisation, validation, and combination of data.
- Create reporting, data analytics and modelling outputs that show year-to-date performance, membership classes, product or membership income, visitation, usage trends, and peak times of use and other reports as required by the fitness and aquatics teams.
- Design the workbook so reporting can be extended to new membership types, product types, or future report structures without requiring specialist technical support.
- Produce clear staff-facing documentation so QUT Sport can refresh, check, and maintain the reporting tool after the student project concludes.

Project Phases

Phase 1: Current State, Data Sources and Reporting Requirements (Weeks 1-4)

- Review the current manual Excel reporting process and understand how reporting outputs are currently produced and checked.
- Identify the recurring raw report exports used by QUT Sport, including membership, financial, visitation, and usage-related reports.
- Document the key metrics required by QUT Sport, including membership categories, product types, income, visits, usage patterns, and peak times.
- Agree the minimum viable reporting outputs and any stretch outputs that would improve business insight.

Phase 2: Excel Reporting Design and Data Preparation (Weeks 5-8)

- Design the workbook structure, including input tabs, transformed data tables, validation checks, master mapping tables, and reporting outputs.
- Pilot models and data analytics for reporting to address project objectives.
- Develop repeatable Excel-based preparation steps to clean and combine the supplied data exports.
- Build validation checks that compare outputs against known reporting numbers where available.

- Create flexible lookup or mapping logic so new membership and product types can be added with minimal future rework.

Phase 3: Reporting Outputs, Visualisation and Handover (Weeks 9-12)

- Produce practical outputs such as summary tables, trend views, charts, filters, and slicers within Excel.
- Include reporting views and outputs of models and data analytics for memberships, financial performance, visitation, usage trends, and peak time-of-day analysis.
- Test the workbook with sample/report export data and document known limitations, assumptions, and data quality considerations.
- Prepare a handover pack and user guide for QUT Sport staff, including refresh steps, validation steps, and notes for future enhancement.

Key Considerations

- Usability is critical: QUT Sport staff should be able to refresh and maintain the reporting tool without specialist data science, API, or Power BI expertise.
- The project should prioritise reliable reporting and validation over technical complexity.
- The workbook should make assumptions visible, especially where categories, product names, membership types, or raw report formats may change over time.
- The solution should support academic calendar-driven fluctuations, including semester peaks and troughs.
- The final workbook should be designed for handover, not just demonstration, with clear tabs, labels, instructions, and version notes.

Deliverables

- A functional Microsoft Excel reporting workbook or workbook package that enhances QUT Sport's current manual reporting process.
- Structured data input and preparation approach for supplied raw report exports.
- Reporting, model and data anlytic outputs for memberships, financial performance, visitation, usage trends, and peak-time analysis.
- Validation checks comparing the workbook outputs to existing QUT Sport report numbers where available.
- A documentation pack explaining data inputs, workbook structure, refresh process, validation process, category/mapping logic, and future improvement opportunities.
- A short handover summary identifying strengths, limitations, assumptions, and recommended next enhancements.